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ZebraSeek integrates complementary facial and clinical evidence for rare disease prioritization

Naoya Yoshikuwa[1], **Hirokazu Chiba**[2], **Teppei Okazaki**[3], **Jaemoon Shin**[2], **Eisuke Dohi**[4], **Hiroyuki Mishima**[5], **Atsuko Yamaguchi**[3], **Tzung-Chien Hsieh**[6], **Orion Buske**[7], **Susumu Goto**[1,2], **and Toyofumi Fujiwara**[2]

1 Department of Computational Biology and Medical Sciences, Graduate School of Frontier Sciences, The University of Tokyo **2** Database Division for Life Science (DBCLS), BioData Science Initiative (BSI), National Institute of Genetics (NIG), Research Organization of Information and Systems (ROIS) **3** Information and Data Sciences, Graduate School of Information and Data Sciences, Tokyo City University **4** National Institute of Neuroscience, National Center of Neurology and Psychiatry (NCNP) **5** Atomic Bomb Disease Institute, Nagasaki University **6** Institute for Genomic Statistics and Bioinformatics, University Hospital Bonn **7** PhenoTips, Toronto, Ontario, Canada

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Abstract

Rare diseases can remain undiagnosed for years because individual disorders are uncommon and clinically heterogeneous, while relevant clues are distributed across structured phenotypes, facial appearance, prior cases and disease knowledge. Computational tools can use Human Phenotype Ontology (HPO)-encoded findings, genomic data and increasingly diverse medical images, but these modalities are processed by models with different outputs and score scales. ZebraSeek converts clinical and facial inputs into ranked disease candidates with specialist tools, then uses a large language model (LLM) to integrate and externally check those candidates while retaining their sources. Facial-image analysis provides a complementary machine-readable phenotype that can capture patterns not fully represented by manually encoded HPO terms. In 74 literature-derived cases spanning 19 diseases, ZebraSeek placed the recorded diagnosis first in 50 (67.6%) and within five in 59 (79.7%); it recovered 11 diagnoses outside PubCaseFinder's top five, while five diagnoses present in PubCaseFinder's top five were absent from ZebraSeek's final top five. During DBCLS BioHackathon 2026, we expanded analysis to 368 patients, 462 images and 54 disorders to examine how candidate depth affects coverage and downstream candidate count, and to assess the feasibility of case-specific candidate selection.

Rare diseases are difficult to diagnose not only because each condition is uncommon, but also because patients can present with incomplete, age-dependent or atypical combinations of findings. Clinicians may therefore have limited prior experience with the eventual diagnosis, and patients can undergo repeated specialist consultations, testing and misdiagnosis before a diagnosis is reached. This prolonged diagnostic odyssey motivates computational approaches that can connect a patient's presentation with rare diseases that may otherwise be difficult to recognize (Zhao et al., 2026).

The Human Phenotype Ontology (HPO) provides a structured vocabulary for clinical abnormalities and has become a common representation for phenotype-driven rare-disease analysis (Gargano et al., 2024). Several established tools use HPO information in different ways. Phenomizer (Köhler et al., 2009) ranks diseases using semantic similarity between a patient's HPO profile and disease annotations. PubCaseFinder (Fujiwara et al., 2018) uses HPO terms to retrieve phenotype-similar case reports and prioritize diseases. LIRICAL (Robinson et al., 2020) uses likelihood ratios to combine HPO observations with genomic information for interpretable diagnostic ranking. Exomiser (Robinson et al., 2014) combines phenotype similarity with variant-level evidence to prioritize candidate genes and variants. Together, these approaches illustrate the established role of structured phenotype information in rare-disease prioritization and provide a reference point for complementary modalities such as medical imaging.

Medical images provide a different source of phenotype information. Rather than requiring every visible abnormality to be recognized, named and encoded manually, a specialist image model can transform the image directly into a quantitative representation or ranked prediction. GestaltMatcher (Hsieh et al., 2022) maps patient portraits into a facial-phenotype space and can match patients with the same molecular diagnosis, including ultra-rare disorders not represented during training. Eye2Gene (Pontikos et al., 2025) uses multimodal retinal imaging to generate gene-level predictions for inherited retinal diseases. Bone2Gene (Bolmer et al., 2026) is being developed to detect and distinguish rare bone diseases from hand radiographs, a routinely acquired image type in children evaluated for growth and bone-age abnormalities. Together, these systems show that clinically useful phenotypic signal can be extracted from several image modalities rather than only from facial photographs.

Imaging should not be treated as an objective ground truth. Performance can depend on image quality, age, population representation, disease coverage and the composition of the training or reference set. Its practical advantage is narrower: image analysis can provide a quantitative, machine-readable phenotype that is less dependent on manual description and may capture patterns that are subtle, difficult to verbalize or incompletely represented by HPO terms. PEDIA (Hsieh et al., 2019), for example, reported that information contained in frontal photographs extended beyond what clinical terms alone captured.

Previous multimodal studies have already demonstrated the benefit of combining such information. PhenoScore (Dingemans et al., 2023) combines facial analysis with HPO-based phenotypic similarity. GestaltMML (Wu et al., 2026) integrates facial images with demographic and clinical information. PEDIA (Hsieh et al., 2019) combines image-derived, phenotype-similarity and molecular scores and uses a linear support vector machine to produce a unified gene-prioritization score. SHEPHERD (Alsentzer et al., 2025) uses phenotype-aware knowledge-graph representations for several rare-disease diagnostic tasks. These studies establish that multimodal fusion can be effective; an additional systems-level question is how to make such integration scalable when the available specialist models operate on different modalities, return different candidate spaces and use scores that are not calibrated to one another.

PEDIA provides a useful example of predefined score fusion. Its linear support vector machine learns a unified decision rule over predefined per-gene scores (Hsieh et al., 2019). This directly addresses multimodal score fusion. ZebraSeek addresses a different systems problem: independently developed specialist tools can return heterogeneous candidate lists, ranks, scores and metadata, and a downstream system must decide which candidates to preserve, how deeply to explore each source and which hypotheses warrant detailed verification. As the number of specialist image and phenotype tools grows, this candidate-level selection and integration problem becomes increasingly important.

Large language models (LLMs) provide one possible integration layer because they can consider candidate identities, ranks, tool-specific metadata, patient findings and retrieved external evidence without reducing all inputs to a single predefined score. This flexibility should not be interpreted as evidence that LLM integration is inherently more accurate than machine-learning fusion; it introduces dependence on model reasoning, normalization, prompt design and candidate availability. Recent LLM-based systems include DeepRare (Zhao et al., 2026), RareAgents (Chen et al., 2026) and MEDDxAgent (Rose et al., 2025). DeepRare (Zhao et al., 2026) is particularly relevant because it orchestrates more than 40 specialist tools and knowledge sources and returns evidence-linked, traceable reasoning. Its published input formulation accepts free-text phenotypes, structured HPO terms and genetic testing results or variant call format (VCF) data rather than facial photographs.

ZebraSeek focuses on the candidate-level problem created when different modalities are handled by separate specialist tools. Clinical phenotypes, facial images and related information are first converted into ranked disease candidates. These candidate records are then integrated by an LLM, checked against external medical information and reduced to a final ranked differential. Facial evidence is introduced through GestaltMatcher rather than by asking the LLM to

interpret a patient photograph directly. The aim is not to replace the modality-specific models, but to expose a common disease-candidate interface while retaining candidate identity, source tool, within-tool rank or score and later evidence links.

This candidate-level interface is designed to accommodate outputs from heterogeneous specialist tools without requiring them to share a single predefined score representation. It can, in principle, accommodate candidate outputs from HPO-based tools, facial-image models, retinal-image models or other specialist systems while allowing the downstream reasoning stage to consider patient-specific context and external evidence. The claim here is flexibility rather than demonstrated superiority: the present study does not establish that LLM-based integration is more accurate than PEDIA or other trained multimodal models.

The current implementation also exposes a concrete limitation of this architecture. Each component contributes its top five candidates using a fixed heuristic. A disease ranked below five is invisible to all later LLM reasoning and verification, even if the downstream reasoning is strong. Conversely, simply increasing every list to dozens of candidates increases the number of diseases that must be normalized, searched, compared and verified. Because PubCaseFinder and GestaltMatcher operate on different modalities and their scores are not directly interchangeable, the practical problem is not only how to combine them, but also how many candidates from each specialist tool should be allowed to reach the expensive downstream stages.

During DBCLS BioHackathon 2026, we investigated this candidate-selection problem rather than treating the fixed top-five setting as immutable. We expanded the analysis to Phenopacket Store v0.1.27 linked with facial images, covering 368 patients, 462 images and 54 disorders, and asked how much diagnostic coverage is hidden below the top five, how quickly the candidate set grows when deeper ranks are included, and whether selecting the number of candidates on a case-by-case basis could offer a more practical balance. Fixed-depth conditions, two zero-shot LLM conditions and a best-case retrospective reference were used as initial baselines for this question rather than as the final method. In parallel, structured outputs were strengthened to keep candidate identity, source tools and uniform resource locators (URLs) explicit through ranking and verification.

Results

Complementary evidence creates both opportunities and a selection problem

The organizing principle of ZebraSeek is that a useful disease hypothesis need not be supported by every component (Fig. 1). Phenotype matching, facial similarity, semantic retrieval and direct LLM prediction provide different routes to the same disease vocabulary. Their outputs are candidate proposals, not independent measurements of diagnostic probability: tools can share phenotype annotations, literature and underlying knowledge. Agreement can support review, but the number of agreeing tools is not a calibrated measure of certainty.

Integration must address two questions in sequence. First, is the recorded diagnosis available in any component's candidate list? Second, if available, does it remain in the final differential? A correct candidate absent from one tool can expand that tool's coverage, whereas discarding a candidate supplied by another can offset this benefit. Figure 1 presents this conceptual distinction; the following analyses examine it using the existing component and final-output summaries.

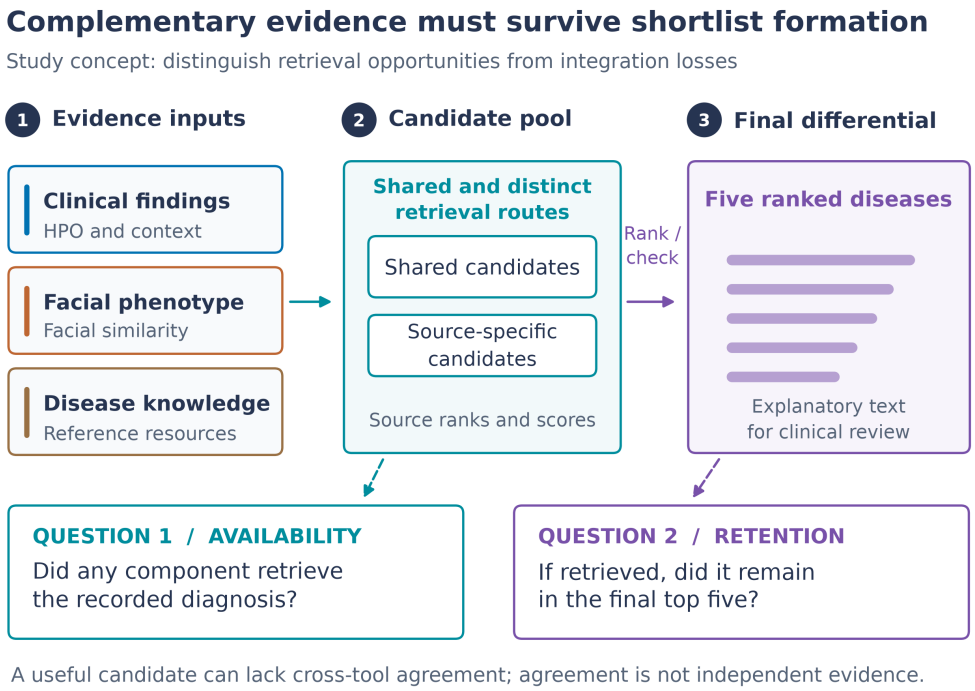


Figure 1: From complementary evidence to a reviewable differential. Different representations can retrieve shared or source-specific disease hypotheses. The conceptual distinction is between candidate availability and retention during shortlist formation. A source-specific candidate may be useful even without cross-tool agreement. Candidate-level integration connects specialist retrieval with LLM ranking and external-information checks; correctness and the value of each stage require separate evaluation.

ZebraSeek integrates candidate proposals from facial and clinical evidence

In the evaluated workflow, ZebraSeek uses a facial image, HPO-encoded clinical findings and recorded sex (Fig. 2). The facial image is processed by GestaltMatcher, while HPO information is supplied to PubCaseFinder and SemanticSearch. The latter retrieves disease candidates by embedding-based similarity using the `text-embedding-3-large` model applied to disease descriptions derived from Mondo. A direct LLM component receives HPO information and sex and separately proposes a differential diagnosis. The evaluated implementation used GPT-5.2 for direct predictions and subsequent integration stages.

Each component returns up to five candidate diseases, yielding at most 20 candidate entries before accounting for overlap. The integration stage compares candidates while retaining information about their source ranks and scores. A verification stage consults external information, including PubMed, and relates candidate diseases to the patient's findings. Explicitly recorded negative findings can be considered during this step. ZebraSeek then returns five ranked candidates with explanatory text. This staged design separates specialist candidate generation from LLM-based integration and external-information checking, while retaining the origin of each candidate for review.

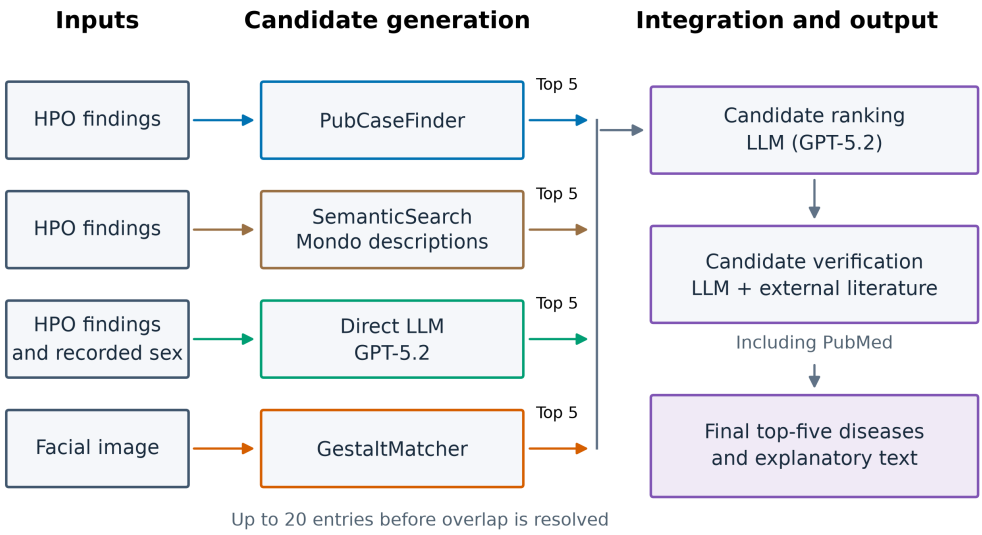


Figure 2: ZebraSeek workflow. Facial images are analyzed by GestaltMatcher; HPO terms are supplied to PubCaseFinder and SemanticSearch; and HPO terms together with recorded sex are supplied to the direct LLM component. Each component provides up to five candidates. LLM-based ranking and literature-based verification produce a final top-five list with explanatory text.

A web interface connects clinical input to staged candidate review

The ZebraSeek input interface supports observed and excluded findings, recorded sex, age of onset, facial-image upload and import from Phenopackets (Fig. 3). It brings structured clinical descriptions and facial evidence into one query workflow.

The results view separates tool output, tentative candidates, a stage labeled validation, and final disease candidates. Component tables display disease identifiers, ranks and similarity scores alongside the final differential. This organization allows users to inspect component proposals as well as the integrated result. The validation label denotes the system's external-information checking stage, not independent clinical validation.

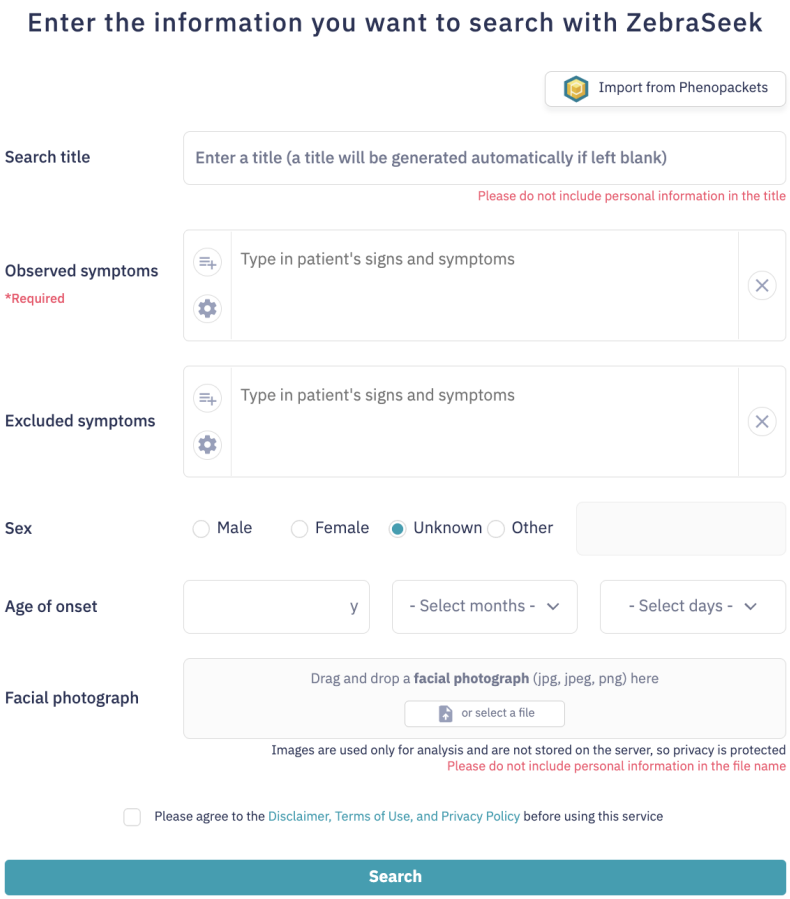


Figure 3: ZebraSeek clinical input interface. The input form supports Phenopacket import, observed and excluded findings, sex, age of onset and facial-image upload. No patient photograph or populated patient record is shown.

Higher observed recall in a benchmark of 74 cases

The evaluation set comprised 74 cases spanning 19 diseases, obtained by matching Phenopacket Store v0.1.25 records to facial images in GestaltMatcher Database. The recorded diagnosis served as the reference label; the stated system inputs were facial images, HPO findings and sex. Top-k recall indicates whether the recorded diagnosis appeared among the first k predictions from each method.

ZebraSeek had the highest observed recall at every reported cutoff (Fig. 4). Recall@1 was 67.6% (50/74), compared with 63.5% (47/74) for PubCaseFinder, 31.1% (23/74) for GestaltMatcher and 18.9% (14/74) for both SemanticSearch and the direct LLM. At rank five, ZebraSeek reached 79.7% (59/74), compared with 71.6% (53/74), 50.0% (37/74), 35.1% (26/74) and 33.8% (25/74) for PubCaseFinder, GestaltMatcher, the direct LLM and SemanticSearch, respectively. Relative to PubCaseFinder, the observed differences were 4.1 percentage points at rank one and 8.1 percentage points at rank five, corresponding to net increases of three and six correctly prioritized cases.

These are descriptive comparisons within the reported benchmark. No confidence intervals, repeated-run variability or significance claims are included in this analysis. The comparison also reflects different input modalities across methods and is not a controlled estimate of the effect of LLM integration alone.

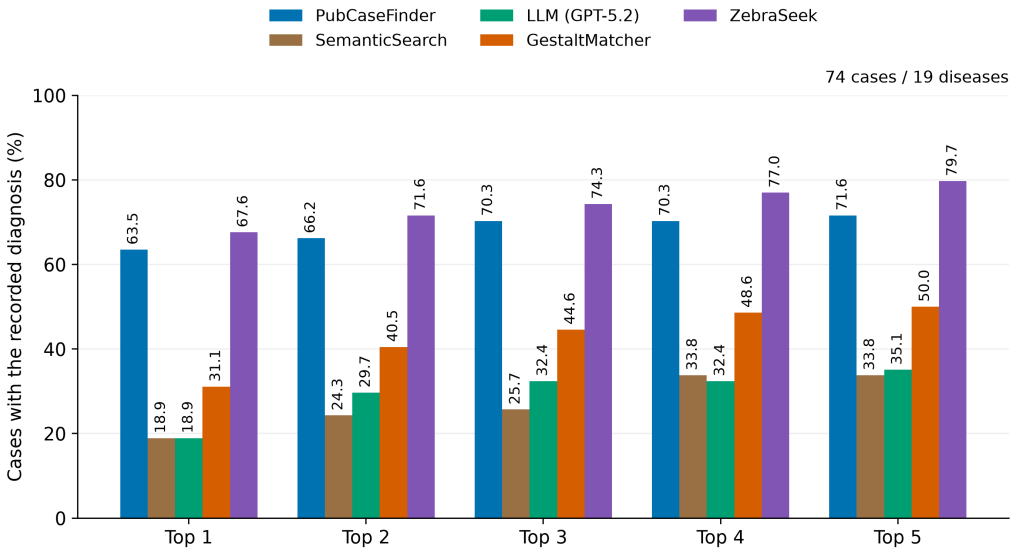


Figure 4: Top-k diagnostic recall across five methods. Top-k recall is the percentage of 74 cases whose recorded diagnosis appears within the first k predictions. The same denominator is used at all cutoffs; this figure reports aggregate results without uncertainty estimates.

Component tools retrieve complementary correct candidates

Among the 59 cases correctly prioritized by ZebraSeek at rank five, the recorded diagnosis was also present in PubCaseFinder's top five for 48 cases, in SemanticSearch's for 25, in the direct LLM's for 26 and in GestaltMatcher's for 35 (Fig. 5). Only 13 of these cases were recovered by all four component tools. The largest exclusive patterns after this four-tool overlap were PubCaseFinder alone (10 cases) and GestaltMatcher alone (seven cases). A further three cases had the recorded diagnosis only in the SemanticSearch candidate list.

ZebraSeek therefore recovered 11 cases in which PubCaseFinder did not retrieve the diagnosis within its top five. Conversely, five cases recovered by PubCaseFinder were missed by ZebraSeek. The difference between these two discordant groups accounts for the net gain of six cases over PubCaseFinder at rank five. This distinction matters: the final list extended coverage beyond the strongest individual component while also losing some of that component's correct candidates.

The observed overlaps support complementary candidate availability, but do not identify the causal contribution of any individual modality. For example, the seven successful cases with GestaltMatcher-only correct candidates motivate a controlled comparison without facial analysis; they do not substitute for that comparison. The integration process may also change its ranking when any component is removed.

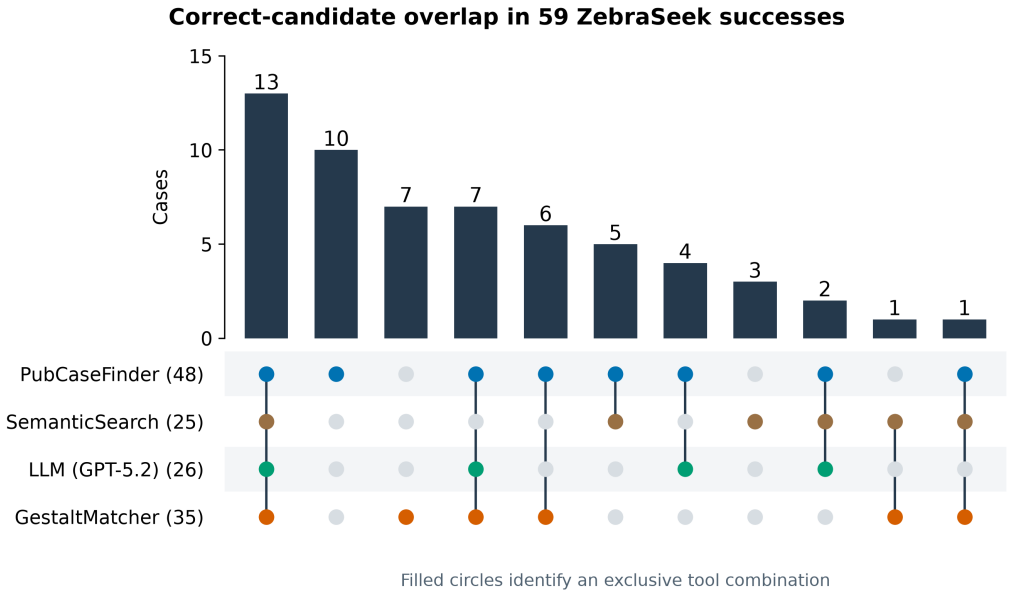


Figure 5: Component overlap among the 59 ZebraSeek successes at rank five. Each column represents an exclusive combination of component tools whose top-five lists contained the recorded diagnosis. Connected filled circles identify the tools in each combination; gray circles indicate absence. Bar heights show case counts. The totals at left are conditional on ZebraSeek success and are not the full-cohort recalls of the component tools.

Candidate coverage and retention account for different failures

ZebraSeek did not include the recorded diagnosis in its top five for 15 cases (Fig. 6). In eight, none of the four component top-five lists contained the diagnosis. These cases expose a coverage limitation in the candidate lists supplied to integration. In the remaining seven, the diagnosis was available from a component but absent from the final list: five had a correct PubCaseFinder-only candidate and two had a correct GestaltMatcher-only candidate. All seven losses therefore involved a diagnosis retrieved by only one component. The aggregate results do not locate the loss within normalization, initial ranking, verification or final selection.

Across the reported overlap patterns, at least one component contained the correct diagnosis in 66 of 74 cases (89.2%). This union is a descriptive measure of candidate coverage with up to 20 entries per case, not a top-five method or a validated performance target. Its difference from ZebraSeek's 59 successful cases identifies seven potentially recoverable diagnoses already present in the inputs.

The 15 failures were concentrated in two of the 19 disease labels: cardiac, facial, and digital anomalies (Online Mendelian Inheritance in Man (OMIM) identifier 618164; 10 cases) and a label abbreviated as neurodevelopmental disorder with coarse facies (OMIM identifier 618505; five cases). This concentration motivates disease-stratified evaluation in addition to case-level averages.

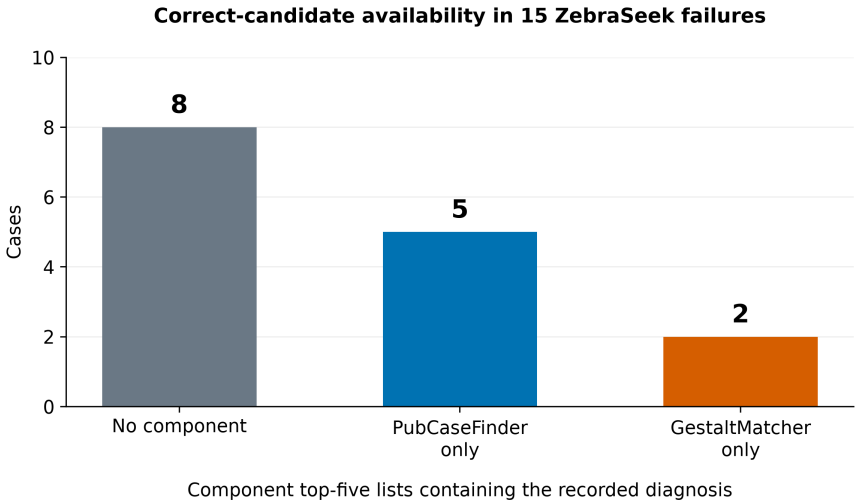


Figure 6: Correct-candidate availability among the 15 ZebraSeek failures at rank five. Eight cases had no correct candidate in any component's top-five list. Five had a correct candidate only in PubCaseFinder and two only in GestaltMatcher. The latter seven cases distinguish failure to retain an available diagnosis from absence of that diagnosis in all initial lists.

Fixed top-five component inputs create a coverage ceiling

The current ZebraSeek workflow receives only the first five candidates from each component. Because later ranking and verification cannot reconsider a disease that was never passed downstream, the fixed input depth creates a limitation distinct from the retention failures described above. To measure how much potentially useful information lies below this cutoff, we performed a hackathon analysis using Phenopacket Store v0.1.27 linked to facial images, comprising 368 patients, 462 images and 54 recorded disorders. This analysis measures candidate coverage at the image level rather than end-to-end ZebraSeek performance; some patients contribute multiple facial images acquired at different ages or time points while sharing the same patient-level phenotype annotations.

PubCaseFinder covered 73.4% of the 462 image-level instances at rank five, 78.1% at rank ten and 83.5% at rank 30. GestaltMatcher covered 30.3%, 34.6% and 45.0%, respectively. After duplicate diseases were removed, the combined PubCaseFinder and GestaltMatcher lists increased from 77.5% coverage at rank five to 82.7% at rank ten and 89.0% at rank 30 (Fig. 7). The important result is not the top-30 number alone, but the gap between shallow and deeper candidate lists: a fixed top-five input prevents later ZebraSeek stages from seeing diagnoses that the specialist tools can retrieve at lower ranks.

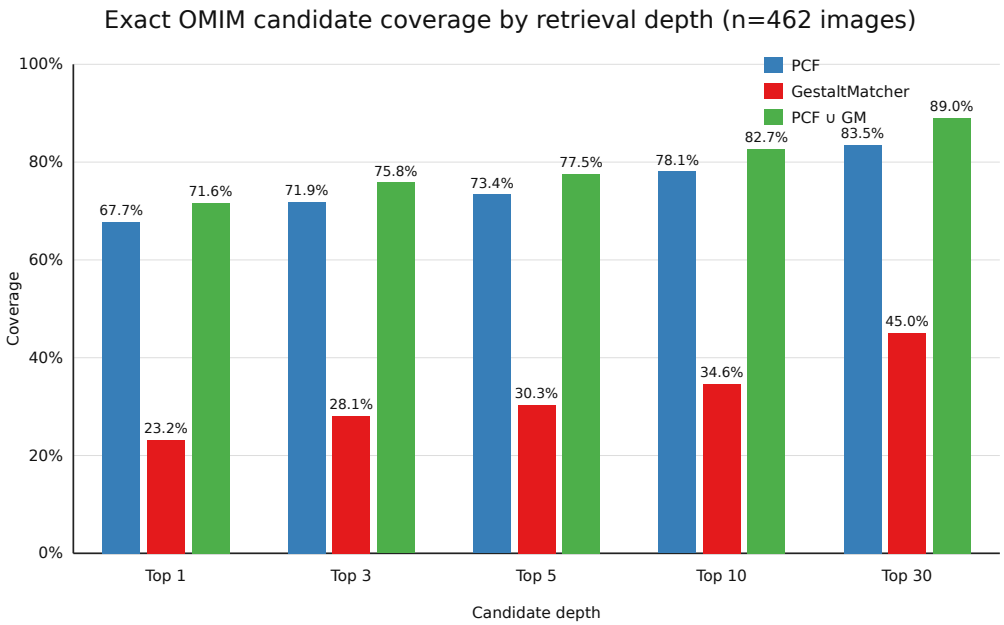


Figure 7: Deeper specialist rankings expose candidates hidden by a fixed top-five input limit. Exact OMIM inclusion was evaluated for 462 facial-image instances linked to 368 patients across 54 disorders. Bars show candidate coverage for PubCaseFinder (PCF), GestaltMatcher (GM) and their combined lists after duplicate diseases were removed at increasing retrieval depths. Coverage increased from 77.5% at top five to 89.0% at top 30. This figure characterizes the candidate set available before ZebraSeek integration; it is not an end-to-end diagnostic-performance comparison.

Baseline analyses show room for more selective candidate exploration

The increased coverage from deeper rankings creates a second problem. At rank 30, PubCaseFinder and GestaltMatcher together can contribute almost 60 unique diseases per image-level instance. Passing every one of these diseases into external search, normalization, LLM comparison and verification would enlarge the downstream workload. The practical objective is therefore not simply to retrieve as deeply as possible, but to identify enough candidates for each case while avoiding unnecessary downstream processing.

This selection cannot be reduced to one shared numeric threshold because the component tools operate on different modalities and their scores have different meanings and scales. PubCaseFinder uses HPO information, whereas GestaltMatcher uses facial images. We therefore used simple reference conditions to test whether candidate coverage and candidate count are meaningfully affected by how many candidates are taken from each tool.

Fixed candidate depths provide baseline reference points (Table 1). PCF@30 reached 83.55% coverage with 30 candidates on average. The combined PCF@30 and GM@30 lists reached 88.96% coverage but produced 57.57 unique candidates on average after duplicate diseases were removed. Two preliminary zero-shot LLM strategies, using no task-specific training examples, selected how many candidates to take from each tool for each input. The input-only strategy achieved 84.38% coverage with 19.28 candidates on average, whereas the strategy that also received tool results achieved 84.60% with 18.18 candidates on average. In this exploratory selector experiment, GPT-5.2 received the facial image when choosing candidate depths; one image-level instance was rejected under the application programming interface (API) content-safety policy, so these LLM-based conditions include 461 instances. This selector experiment is separate from the main ZebraSeek diagnostic path, in which facial images are interpreted by GestaltMatcher rather than by the LLM. These conditions are treated as feasibility baselines rather than evidence that LLM-based selection is optimal.

As a best-case retrospective reference, we used the known diagnosis to identify the smallest number of candidates that would still include the correct diagnosis whenever it was available within the evaluated rankings. This reference reaches the same 88.96% coverage as taking the top 30 candidates from both PCF and GM while retaining 2.66 candidates on average. Because it uses the known diagnosis, it is not an executable diagnostic method or a performance target. It indicates that the large candidate set produced by exhaustive deep retrieval contains substantial redundancy from the perspective of including the correct diagnosis, motivating development of practical case-specific selection methods.

Table 1 | Candidate coverage and average candidate count in the expanded analysis. The + sign denotes the combined candidate lists after duplicate diseases were removed, not arithmetic addition.

Selection condition	Correct-diagnosis coverage	Mean unique candidates	n
PCF@5	73.38%	5.00	462
PCF@10	78.14%	10.00	462
PCF@30	83.55%	30.00	462
PCF@5 + GM@5	77.49%	9.66	462
PCF@10 + GM@5	81.60%	14.58	462
PCF@30 + GM@30	88.96%	57.57	462
LLM-selected depth, input only	84.38%	19.28	461
LLM-selected depth, input + tool results	84.60%	18.18	461
Minimum-depth reference using known diagnosis	88.96%	2.66	462

The candidate count in Table 1 is a rough measure of how many diseases later stages would need to process, not a direct measurement of computational cost. If PubCaseFinder or GestaltMatcher returns its top 30 in a single request, accepting a shallower depth does not make that upstream tool execution cheaper. The intended saving is in the number of diseases passed to external search, LLM verification and final ranking. Actual token use, latency, API calls and monetary cost remain separate end-to-end evaluation measures.

Hackathon implementation strengthened traceable candidate handling

The hackathon implementation also strengthened the structured-output contract used between candidate generation, ranking and verification. Candidate decisions are tied to stable candidate identifiers rather than unconstrained disease-name generation, and source-information fields are carried with the candidate record. The revised schema additionally requires source references, including uniform resource locators (URLs) where available, so that evidence used during verification can remain attached to its retrieval source.

This change supports ZebraSeek's original goal of a reviewable differential, but it is distinct from candidate-selection performance. A URL field does not by itself establish that a retrieved source exists, supports the associated claim or was interpreted correctly. Citation fidelity and expert review remain separate evaluation targets. The immediate implementation goal is to make candidate identity and source information explicit and less likely to be lost during LLM-mediated transformations.

Discussion

ZebraSeek addresses a practical integration problem that becomes more important as rare-disease diagnosis incorporates more specialized data modalities. HPO-based tools, facial phenotyping, retinal imaging, radiographic analysis and genomic prioritization can each produce useful hypotheses, but their outputs differ in representation, score scale and disease coverage. ZebraSeek therefore treats the disease candidate, together with its source and tool-specific metadata, as a common interface between specialist analysis and downstream reasoning.

Facial phenotyping is the first imaging example in the present implementation rather than the endpoint of the architecture. GestaltMatcher (Hsieh et al., 2022) can encode portraits into a phenotype space without requiring every visible feature to be manually translated into HPO terms. PEDIA (Hsieh et al., 2019) showed that image-derived facial information adds diagnostic signal beyond clinical terms when combined with phenotype and molecular evidence. These findings support the use of facial images as a complementary phenotype source, although image analysis remains sensitive to photograph quality, patient age, population representation and reference-dataset coverage.

The original 74-case benchmark primarily supports the effectiveness of combining complementary candidate sources. ZebraSeek achieved higher observed top-k recall than each individual component within that evaluation setting. More importantly, the case-level pattern explains how that gain arose: 11 diagnoses were recovered beyond PubCaseFinder's top-five coverage, whereas five diagnoses present in PubCaseFinder's top five were absent from ZebraSeek's final top five. This shows both the value and the risk of candidate-level integration. Expanding the candidate pool can recover diagnoses missed by an individual tool, but the downstream ranking process can still discard useful source-specific candidates. The seven successful cases in which GestaltMatcher alone contained the recorded diagnosis within its top five are consistent with facial evidence being complementary, but they are not a controlled facial-image ablation and should not be interpreted causally.

Existing multimodal methods demonstrate that combining modalities can be highly effective. PhenoScore (Dingemans et al., 2023) integrates facial and HPO-based information, and GestaltMML (Wu et al., 2026) combines facial images with demographic and clinical information. PEDIA (Hsieh et al., 2019) uses a linear support vector machine to combine predefined image, phenotype and molecular scores. PEDIA addresses multimodal score fusion, whereas ZebraSeek addresses a different systems problem: deciding which heterogeneous specialist outputs should be explored, retained and verified for an individual patient. ZebraSeek's candidate records remain identifiable and can be reconsidered together with within-tool ranks or scores, observed and excluded findings and retrieved medical evidence. This is a flexibility claim, not a demonstrated accuracy advantage over trained fusion models.

DeepRare (Zhao et al., 2026) shows that LLM-based rare-disease diagnosis can orchestrate many tools and knowledge sources while returning reasoning linked to verifiable evidence. RareAgents (Chen et al., 2026) and MEDDxAgent (Rose et al., 2025) likewise demonstrate agentic or modular diagnostic workflows. ZebraSeek should therefore not be positioned as the first agentic or traceable system. Its narrower contribution is to make modality-specific specialist outputs explicit at the candidate level and to treat the number of candidates passed from each specialist tool into downstream LLM reasoning as a design variable. DeepRare's published input formulation includes free-text clinical descriptions, HPO terms and genetic testing results or VCF data rather than facial photographs (Zhao et al., 2026), leaving image-specialist outputs as one area where a candidate-level interface can be useful.

That design variable matters because the current ZebraSeek implementation uses a heuristic top-five cutoff for every component. This choice keeps the integration input small, but it creates an upper limit on what later reasoning can recover. A stronger LLM cannot select a disease that was never passed to it. At the opposite extreme, forwarding dozens of candidates from every tool increases the amount of evidence retrieval, comparison and LLM-based verification.

The system is therefore affected both by the quality of LLM reasoning and by the upstream decision about which candidate diseases the LLM is allowed to consider.

The expanded BioHackathon analysis makes this trade-off measurable. Deeper PCF and GM rankings contain correct diagnoses that are not present in their top-five lists, but exhaustive PCF@30 plus GM@30 produces 57.57 unique candidates on average. The fixed-depth and zero-shot comparisons were not intended to identify a winning method; they were first checks of whether case-specific candidate selection is a worthwhile direction. GPT-5.2 was used here as an exploratory feasibility probe rather than as the intended long-term selector. Future versions are intended to use a local LLM or another locally deployed selector so that this decision can be made without relying on an external image-processing API. The best-case retrospective reference further shows that, in principle, far fewer candidates would have been sufficient when the correct depth is known in hindsight. The next research step is therefore to develop practical selection rules that approach broad coverage without passing the full deep ranking to later stages.

The broader motivation extends beyond facial photographs. Eye2Gene (Pontikos et al., 2025) uses fundus autofluorescence, infrared imaging and optical coherence tomography to generate gene-level predictions for inherited retinal diseases; in a subset of 130 individuals, its image-based rank was higher than or equal to an HPO-based Exomiser-hiPHIVE rank in more than 75% of patients. Bone2Gene (Bolmer et al., 2026) applies deep learning to hand radiographs for rare bone-disease detection and differential diagnosis, illustrating how routinely acquired skeletal images may become another specialist phenotype source. These tools support a future architecture in which facial, retinal, skeletal and structured clinical phenotypes are not forced into one raw representation. Instead, each specialist model can produce ranked candidates that are mapped to a common disease or gene space and then selectively carried forward for deeper verification.

This extension sharpens the efficiency problem. As more modalities are introduced, forwarding every candidate from every specialist tool to the LLM becomes less attractive. A scalable system will need to determine not only how to combine heterogeneous evidence, but also which modality-specific outputs are informative enough for a given patient to justify additional retrieval and reasoning. The BioHackathon candidate-depth analysis is an initial step toward that broader goal because it isolates one tractable version of the problem: how far to explore ranked outputs from HPO-based and facial-image-based specialist tools before downstream integration.

The structured-output revision addresses a separate but complementary weakness of LLM-mediated pipelines. Free-form transformations can change disease names, lose identifiers or detach evidence from the candidate it was meant to support. ZebraSeek therefore uses stable candidate identifiers and carries source fields, including URLs where available, through later stages. This improves source traceability, but it should not be confused with proof of evidence quality. Whether a URL exists, whether the source actually supports the generated statement and whether the medical interpretation is correct require separate validation.

Several limitations constrain the present interpretation. The original benchmark is retrospective, small and selected for availability of both phenopackets and facial images. Facial-image models can be sensitive to photograph quality, patient age, population representation and the composition of their reference data. The expanded analysis contains 462 images from 368 patients; some patients therefore contribute multiple image-level observations from different ages or time points while sharing the same phenotype annotations, so these observations are not independent patient-level tests. The reported 54 disorders remain unevenly represented. In the zero-shot LLM analysis, one image was rejected by the API content-safety policy, reducing the denominator from 462 to 461. No controlled end-to-end comparison has yet shown that case-specific candidate selection improves final ZebraSeek Recall@5, latency, token use or monetary cost. Repeated-run variability, disease-stratified performance and patient-wise held-out evaluation are also required.

The evaluation cases used for GestaltMatcher were excluded from its training data, reducing direct training overlap for the facial-matching component. The source case reports were not directly supplied to ZebraSeek as query inputs. However, because those reports are publicly available, prior exposure through LLM pretraining or other indexed resources cannot be excluded, and downstream retrieval may encounter the same or related publications. Independent cases remain necessary to test generalization beyond these potential routes of information overlap. Expert evaluation is additionally required to assess explanations and any effect on diagnostic decisions. The present findings do not establish molecular diagnostic yield, clinical benefit or equitable performance.

The resulting research question is therefore not simply whether more modalities or a stronger LLM improve accuracy. It is how to acquire complementary specialist candidates from very different data types, select them narrowly enough to keep downstream reasoning practical, combine them flexibly without losing their identity, and preserve enough source information for the final differential to be independently reviewed. Facial phenotyping provides the first multimodal test case in ZebraSeek, while tools such as Eye2Gene and Bone2Gene illustrate the wider direction toward efficient integration of multiple specialist imaging modalities. The current ZebraSeek implementation, original 74-case benchmark and BioHackathon candidate-depth analysis provide an initial framework for that broader problem rather than a completed clinical validation.

Methods

Study design and original case selection

The original study retrospectively evaluated 74 literature-derived cases covering 19 diseases. Phenotypic descriptions, sex and recorded diagnoses were obtained from Phenopacket Store v0.1.25, a corpus built using the Global Alliance for Genomics and Health (GA4GH) Phenopacket representation (Danis et al., 2025). Cases were linked to facial images in GestaltMatcher Database, a resource for facial phenotyping (Hsieh et al., 2024), by matching records corresponding to the same individual across the two resources. The evaluation cases used for GestaltMatcher were excluded from its training data. The source case reports were not directly supplied to ZebraSeek as query inputs; however, because the reports are publicly available, prior exposure through LLM pretraining or other indexed resources cannot be excluded.

Expanded analysis of candidate-list depth

The hackathon analysis used Phenopacket Store v0.1.27 and matched facial-image data, yielding 368 patients, 462 facial-image instances and 54 recorded disorders. Some patients contributed multiple facial images acquired at different ages or time points. These image-level instances shared the same patient-level phenotype annotations, while the facial image and the age corresponding to that image could differ. Candidate coverage was therefore evaluated at the image-instance level. PubCaseFinder and GestaltMatcher rankings were examined to a maximum depth of 30. Exact-OMIM coverage was defined as the fraction of evaluated instances for which the recorded OMIM diagnosis was present in the accepted candidate set after duplicate diseases were removed.

PubCaseFinder uses HPO information, whereas GestaltMatcher uses facial images. Their internal scores may have different meanings and scales and are not assumed to be directly comparable. The common information available to downstream ZebraSeek integration is therefore the ranked disease candidate, together with tool-specific rank, score and source metadata.

For an instance x_i with recorded reference diagnosis y_i, let $R_j(x_i) = (c_{ij1}, \ldots, c_{ijK_j})$ be the ranking returned by candidate generator T_j. A selection strategy chooses an accepted depth k_{ij} for each generator. The downstream candidate set is

$$C_\pi(x_i) = \bigcup_j \{c_{ij1}, \ldots, c_{ijk_{ij}}\},$$

with duplicate diseases removed. Coverage is

$$\text{Coverage}(\pi) = \frac{1}{N} \sum_i \mathbf{1}[y_i \in C_\pi(x_i)].$$

The practical objective is to retain high candidate coverage while reducing the mean number of unique diseases passed to later stages, or ultimately the measured downstream computational cost. Candidate count and actual compute cost are reported separately because the two are not equivalent. In the present implementation PubCaseFinder and GestaltMatcher can return deep rankings in a single execution, so the accepted depth primarily changes the number of candidates entering later search, LLM verification and ranking stages rather than the cost of the upstream tools themselves.

Candidate-selection comparison conditions

Fixed-depth baselines applied the same accepted depth to all instances. The reported conditions include PCF@5, PCF@10, PCF@30, and selected PCF–GM combinations with duplicate diseases removed. Mean candidate count was calculated after deduplication across the two component rankings.

The zero-shot LLM conditions used GPT-5.2 through the API with reasoning effort set to medium and no task-specific training examples. The LLM was asked to choose candidate depths per input; one condition used input information only, and a second also received component-tool results. In this exploratory selector experiment, GPT-5.2 received the facial image for depth selection. This differs from the main ZebraSeek diagnostic architecture, where facial images are interpreted by GestaltMatcher rather than by the LLM. One image-level instance could not be evaluated because the image was rejected under the API content-safety policy, so aggregate coverage and mean candidate counts are reported for 461 image-level instances. GPT-5.2 was used here to test feasibility; a future selector is intended to use a local LLM or another locally deployed model.

For a best-case retrospective reference, the known diagnosis and its position in each ranking were used to choose the smallest accepted candidate set that still contained the correct diagnosis whenever it was available within the maximum evaluated depths. This reference is used only to estimate how much the number of downstream candidates could theoretically be reduced if the appropriate depth were known in advance; it is not an executable diagnostic method.

Input preparation and reference diagnoses

ZebraSeek used facial images, HPO findings and recorded sex. The reference diagnosis was used for evaluation rather than intentionally supplied as a model input. HPO provides standardized identifiers for phenotypic abnormalities (Gargano et al., 2024). The workflow can also consider explicitly recorded negative findings.

Interface documentation

The application interface supports Phenopacket import, observed and excluded findings, recorded sex, age of onset and facial-image upload. The results interface exposes component outputs, tentative candidates, external-information checking and final disease candidates.

Candidate generation

PubCaseFinder generated candidates from HPO findings using phenotype-based disease matching (Fujiwara et al., 2018). GestaltMatcher generated candidates from facial images (Hsieh et al., 2022). SemanticSearch compared phenotype information with embeddings generated by `text-embedding-3-large` for disease descriptions obtained from Mondo, which integrates disease terminology across resources (Vasilevsky et al., 2026). The direct LLM component used HPO findings and sex in a zero-shot setting. Each component supplied up to five candidates in the original ZebraSeek benchmark. The expanded analysis separately examined PubCaseFinder and GestaltMatcher to rank 30.

Candidate integration, verification and source tracking

The original integration stage received up to 20 candidate entries with source ranks and scores and used an LLM to rank candidate diseases. A verification stage related candidates to the available phenotype information and consulted external medical information, including PubMed. The workflow then generated a final top-five ranking and explanatory text.

During the hackathon the structured-output contract was strengthened to preserve candidate identity and source information through these stages. Candidate decisions are represented with candidate identifiers rather than relying only on unconstrained disease-name text, and source-reference fields include URLs where available. This structure is intended to make the origin of evidence and candidate transformations easier to audit; factual correctness and citation fidelity remain separate evaluation targets.

Evaluation metrics and descriptive analysis

For the original benchmark, top-k recall was the fraction of 74 cases for which the recorded diagnosis occurred among the first k predictions, with k ranging from one to five. The expanded depth-selection analysis instead uses candidate coverage: whether the exact reference OMIM diagnosis appears anywhere in the accepted candidate set after duplicate diseases are removed. These quantities answer different questions and are not directly interchangeable.

The overlap summaries distinguish successful ZebraSeek cases from its failures and use the component top-five lists. Candidate availability denotes presence of the recorded diagnosis in at least one component top-five list; retention denotes its continued presence in the final top-five list when initially available.

The expanded analysis reports candidate coverage and mean unique downstream candidate count as separate axes. Candidate count is treated as a rough indicator of how many diseases later stages must process rather than as a direct measurement of API calls, token use, wall-clock time or monetary cost.

Ethics and data governance

The benchmarks are secondary analyses of literature-derived cases and associated facial images. This manuscript reports aggregate figures, schematics and an unpopulated interface screenshot and does not redistribute patient photographs or populated patient records.

Data availability

The aggregate values used for the original figures are provided in `paper/data/aggregate_results.json`. The hackathon candidate-depth summary is provided in `paper/data/adaptive_candidate_depth_summary.csv`. These files contain aggregate evaluation values rather than patient-level source data. Phenopacket Store is described in the cited resource paper (Danis et al., 2025). Facial-image access is governed by GestaltMatcher Database. The manuscript repository does not redistribute the underlying patient images.

Code availability

The manuscript source and scripts used to reproduce its aggregate figures are available at the manuscript repository. The ZebraSeek implementation is developed at 708san/AI_AgentWith-LangGraph, whose public README describes the rare-disease diagnosis pipeline, Phenopacket runner and optional prompt/node logging. The manuscript figure-generation scripts reproduce aggregate study summaries; they are not the diagnostic implementation.

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